

JRU Form 30
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JOSE RIZAL UNIVERSITY

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MANDALUYONG CITY

EXAMINATION BOOK

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SUBJECT ITC C506-3021 ELECTIVE

DO NOT TEAR ANY PAGE OUT OF THIS
EXAMINATION BOOK

TEST I

1) Predicting Delay-Risk will be more useful to the DOE, since it's categorical unlike travel distance which is numerical. Being categorical means we can use Logistic Regression to have a much more accurate output to present to DOE. Additionally, presenting the outputs for Delay-Risk will be much more beneficial to DOE, since they are asking for a model that can determine which scheduled delivery is at "Critical Risk" of Delay and using logistic Regression to predict the outputs of Delay-Risk, they will have an easier way to understand which delivery is at "Critical Risk" of Delay.

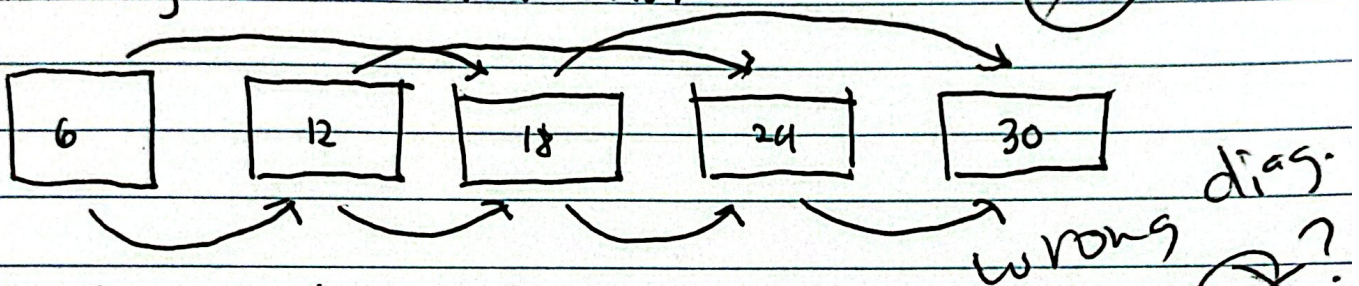
2) I will have to use logistic Regression, since its characteristics are for categorical. It can use Sigmoid Curve to determine which of the scheduled delivery is at "Critical Risk" of Delay. It can also use Confusion Matrix as a way to present which delivery is at Risk of Delay.

3) If the model outputs 0.34 for TAK-205, then DOE will be in luck because one of their scheduled delivery is at low Risk of Delay. We can determine this by using Sigmoid Curve and putting the threshold on 0.5. Since the model outputs 0.34, it never crossed the threshold meaning it's at low Risk. We can also do simple probability conversion, since the model outputs 0.34, we can round that up to 34% and since probability is only 0% - 100%. Having a chance of Delay of 34% out of 100% is a good result since it's saying it has a low chance of Delay for the delivery.

TEST II

1) Having only a test set containing 12 rows will give a false sense of accuracy. Why? Because the 12 rows only contain "low Risk" deliveries and by only having that the model will be highly optimistic to low risk deliveries since that is the only data it knows.

2) To have a more realistic and accurate outputs we will be using 6-Fold Cross-Validation.



This diagram shows that the 30 rows is grouped into 5 boxes as a mini version to present how cross-validation works. First, it checks every data (arrows below the box). Then, it checks the data into segments (arrows on top of the box) to validate if it is using every data. Cross-validation is a way to present data that is realistic and accurate because it is using every part of the dataset unlike ~~the~~ Train-Test Split that can be optimistic on the testing part.

3) If the DOE relies on this model the False Negative it can get is to dispatch the limited tankers to a province that is not in need of the limited tankers. And that can cause a delay to those in need of the tankers.